Ex.No:11 **Date:07/10/25**

EXAMINING NETWORK ADDRESS TRANSLATION (NAT) USING CISCO PACKET TRACER

Aim:

To Examine Network Address Translation (NAT) using Cisco Packet Tracer.

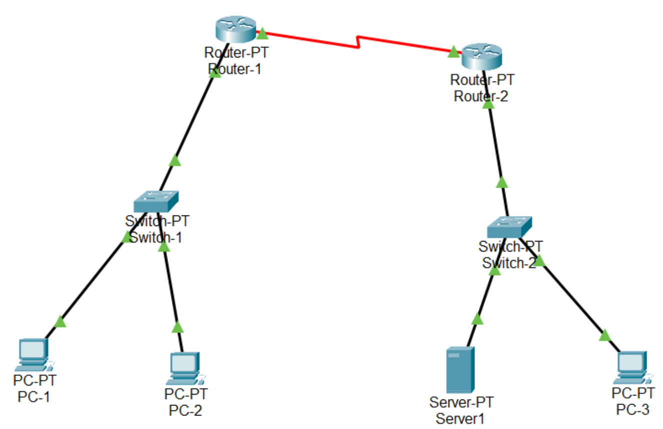
1. Setting Up the Network Topology

Devices Required:

- 2 Routers (R1, R2)
- 2 Switches (S1, S2)
- 3 PCs (PC1, PC2, PC3)
- 1 Server
- Serial and Ethernet cables

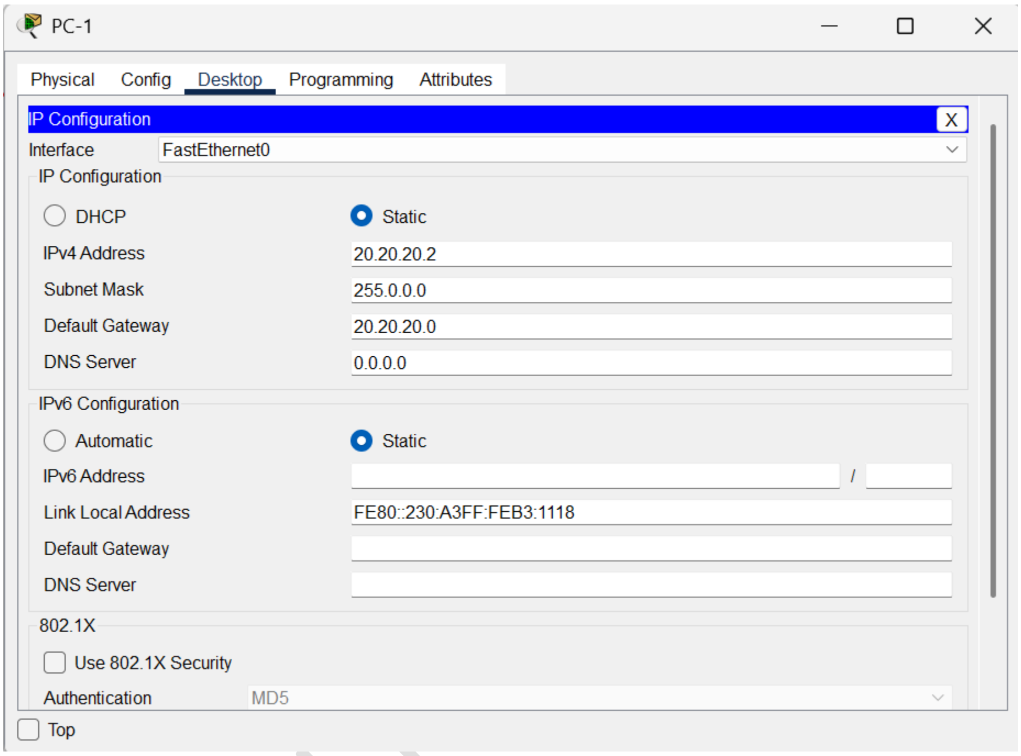
Cable the devices:

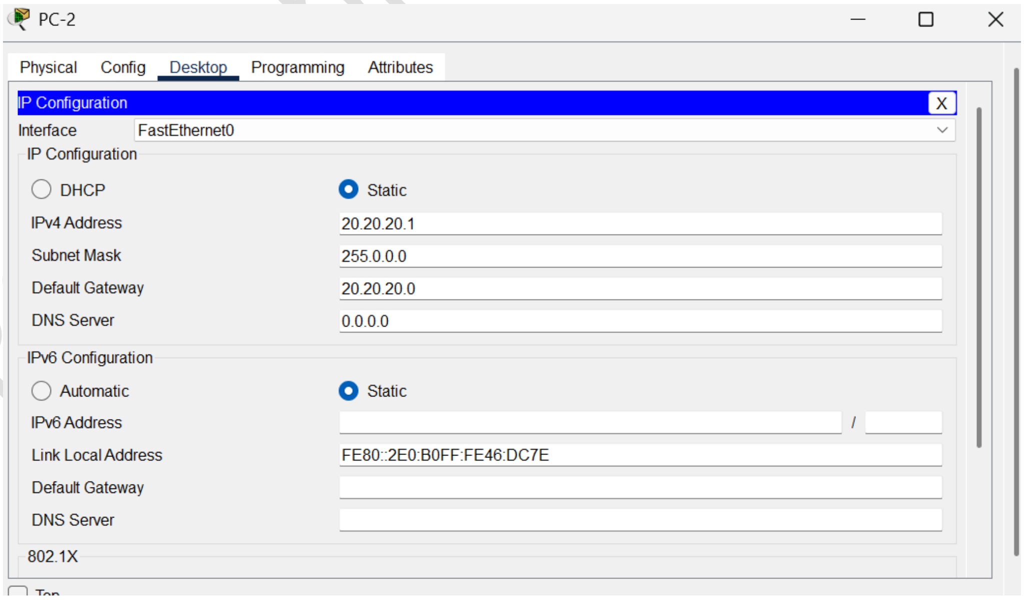
- PC1 → FastEthernet0 → Switch S1.
- PC2 → FastEthernet0 → Switch S1.
- Switch S1 → GigabitEthernet0/0 → Router R1.
- Server → FastEthernet0 → Switch S2.
- Switch S2 → GigabitEthernet0/0 → Router R2.
- Router R1 Serial0/0/0 ↔ Serial0/0/0 Router R2 (serial cable).

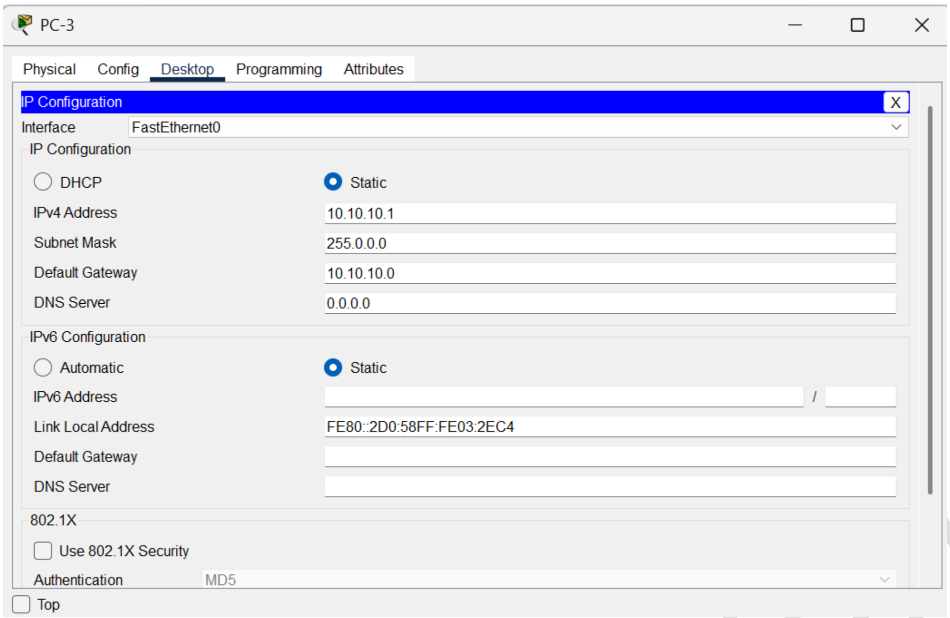


2.Assign IP Addresses to PC and Server:

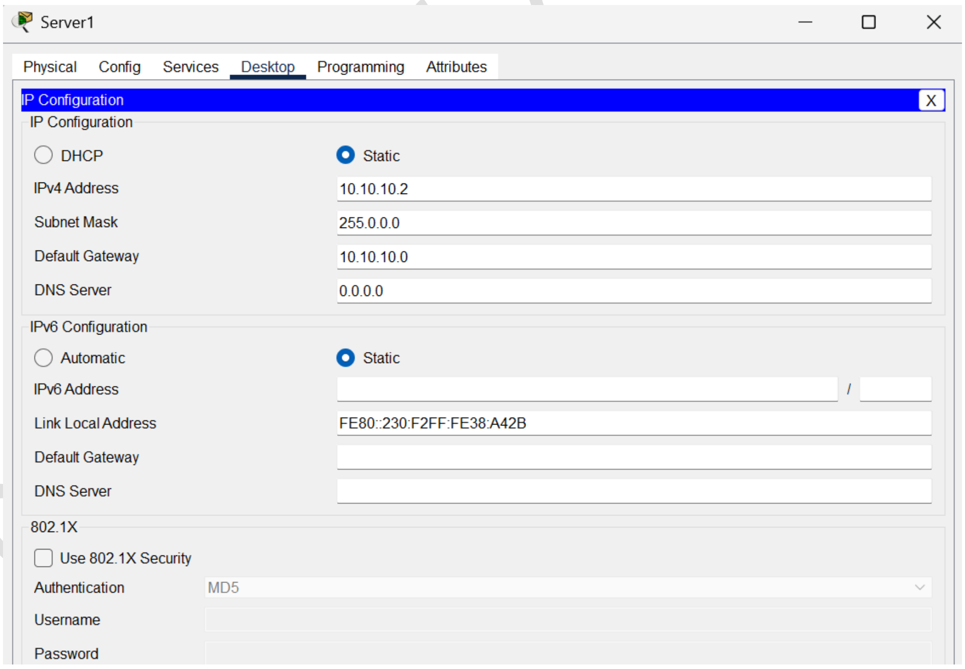
1. Open the PC.
2. Go to Desktop → IP Configuration.
3. Enter the details:



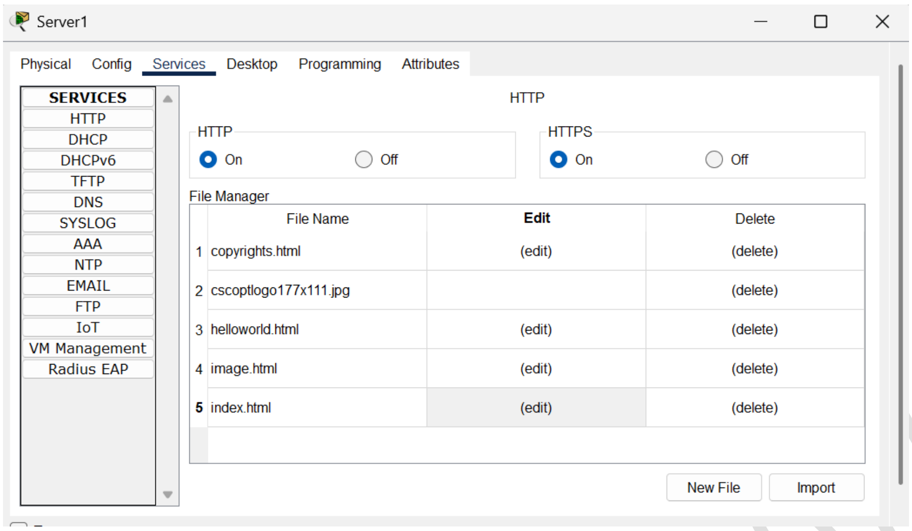




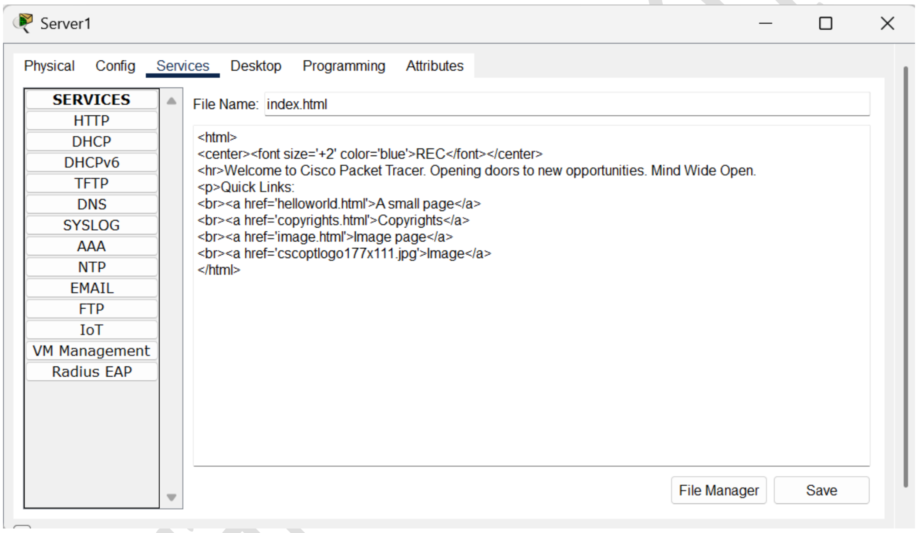
1. Open the Server.
2. Go to Desktop → IP Configuration.
3. Enter the details:



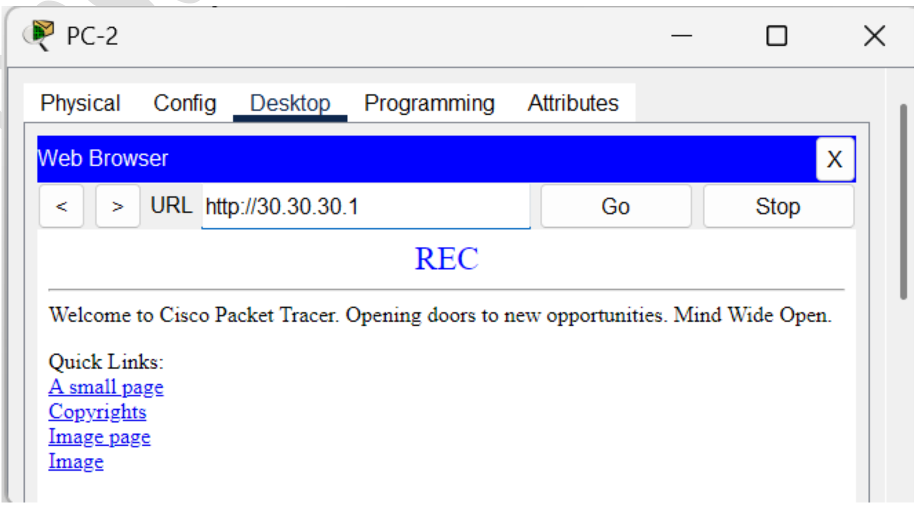
1.Go to services tab,edit index.html



2.Change the heading.



3.Open browser in PC1 and type the following url. Notice the Change in heading.



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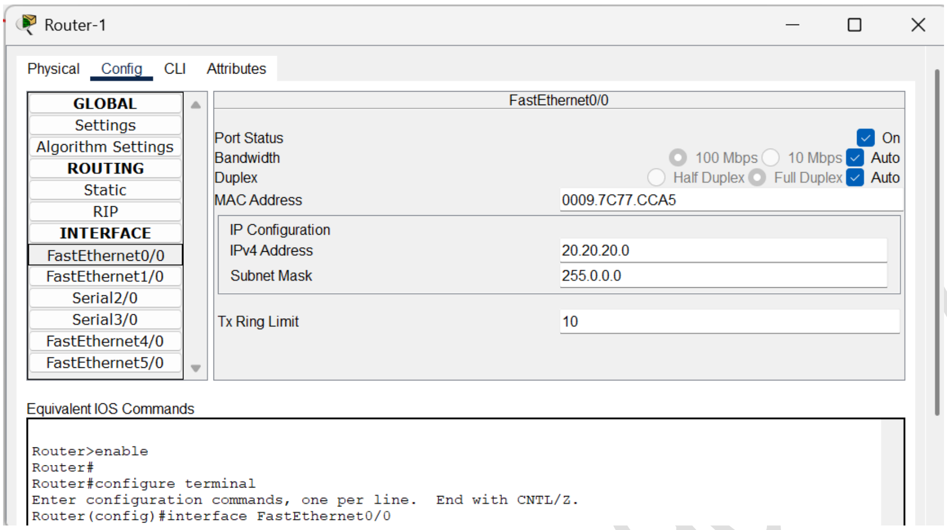
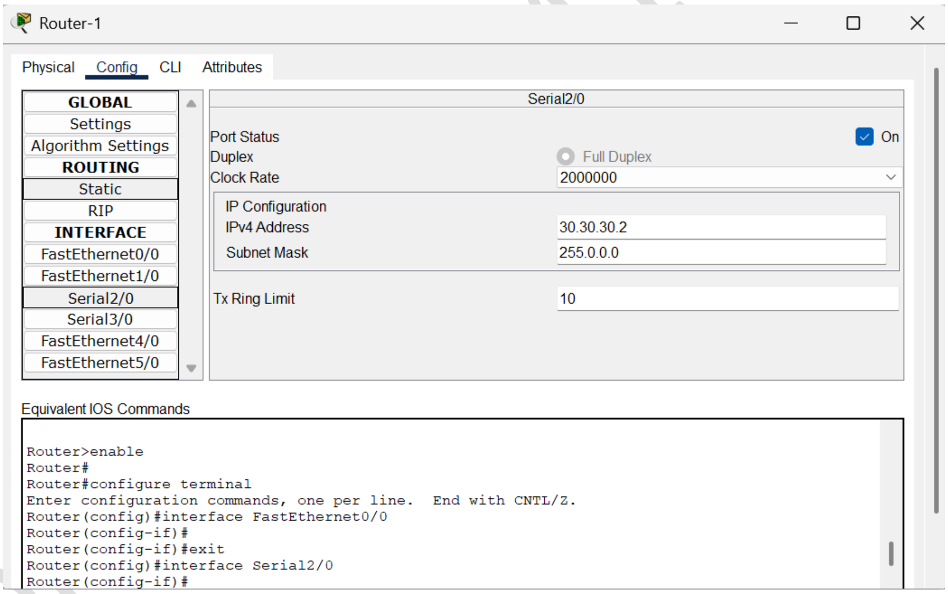
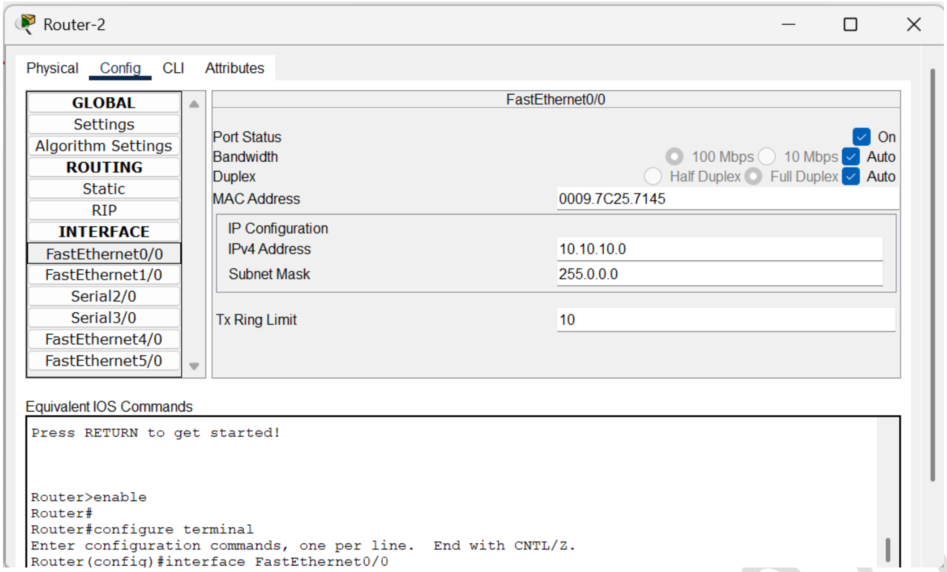
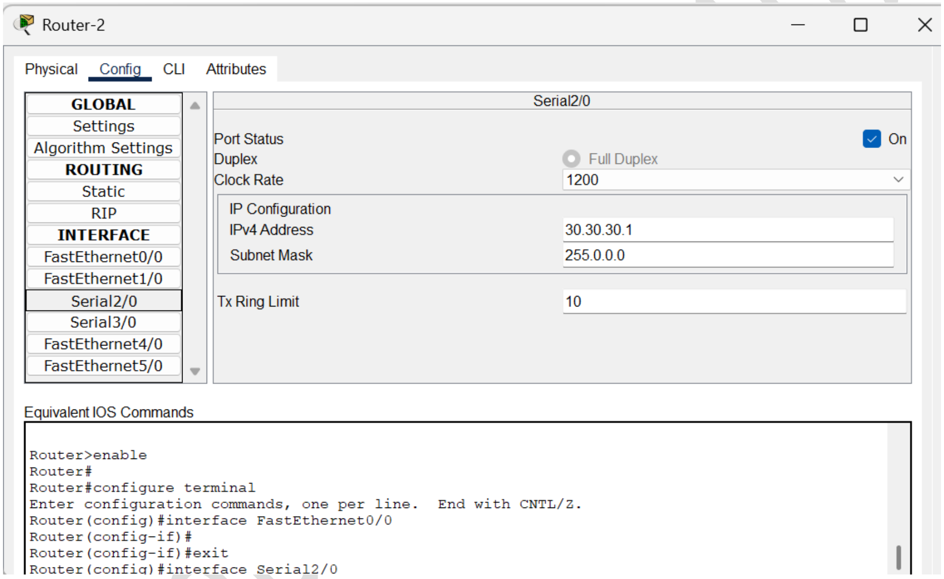
3.Configure the routers

- Click on the router,go to config tab.
- Under interface,select the port connected . Mark the tick box “on”. Fill the ip address and subnet mask.
- Repeat for all the ports in both the routers.









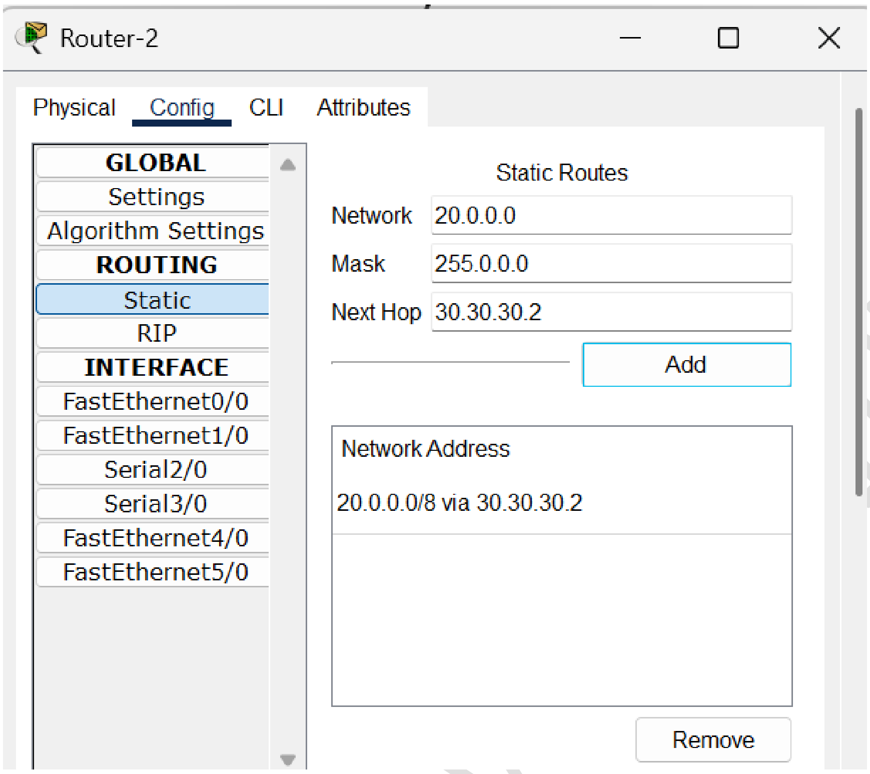
Static NAT Mapping on Router1:

- Open R1 → CLI
- Enter:

```
Router(config)#ip nat inside source static 10.10.10.1 30.30.30.1
Router(config)#ip nat inside source static 10.10.10.2 30.30.30.1
Router(config)#int f0/0
Router(config-if)#ip nat inside
Router(config-if)#int s2/0
Router(config-if)#ip nat outside
Router(config-if)#exit
```

Static Routing on Router2:

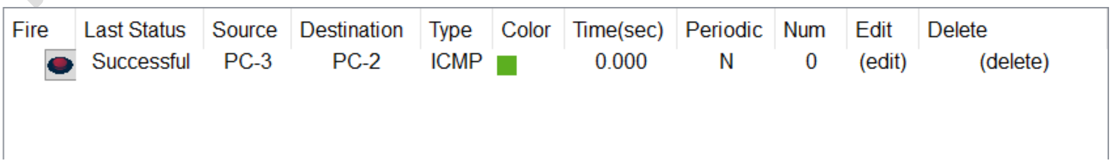
- Open Router2 → Config tab → Static.
- Fill the fields: Network,Subnet Mask,Next Hop
- Click Add



4.Test Connectivity Using PDU and Ping

PDU Test:

- Select the Add Simple PDU tool from the main toolbar.
- Source: PC3 → Destination: PC2.
- Send the PDU.
- The PDU suceeds because the private IP can traverse the public network.



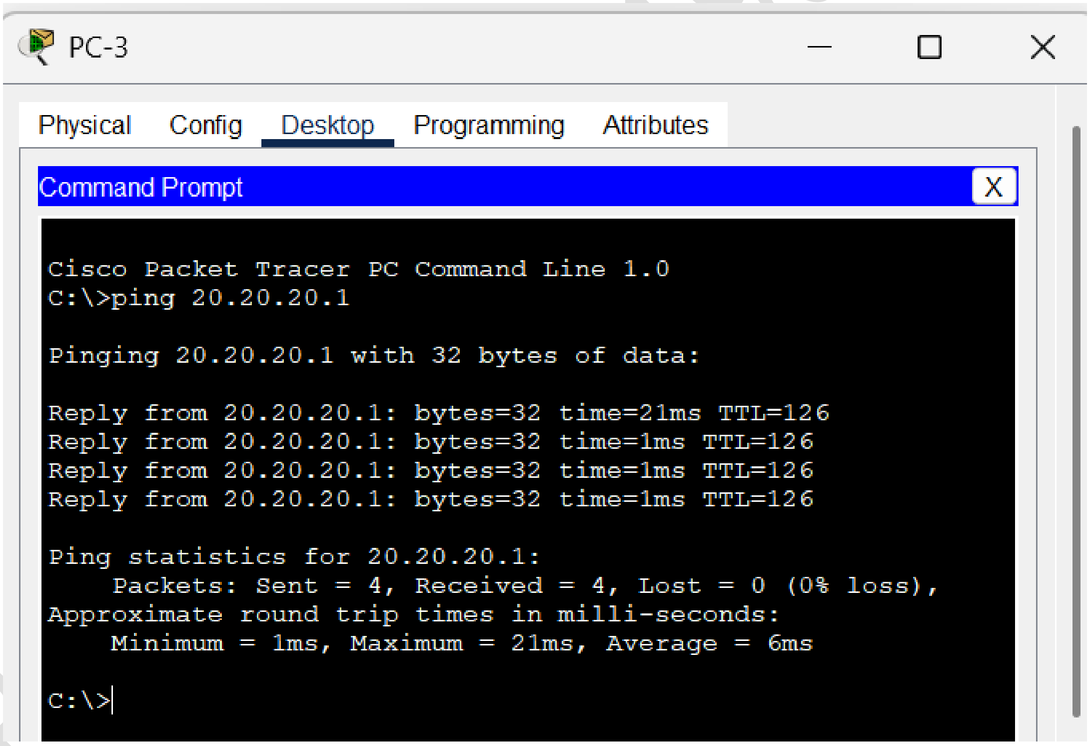
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC-3	PC-2	ICMP		0.000	N	0	(edit)	(delete)

- Select the Add Simple PDU tool from the main toolbar.
- Source: PC2 → Destination: PC3.
- Send the PDU.
- The PDU fails because the public IP cannot traverse the private network.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC-3	PC-2	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC-2	PC-3	ICMP		0.000	N	1	(edit)	(delete)

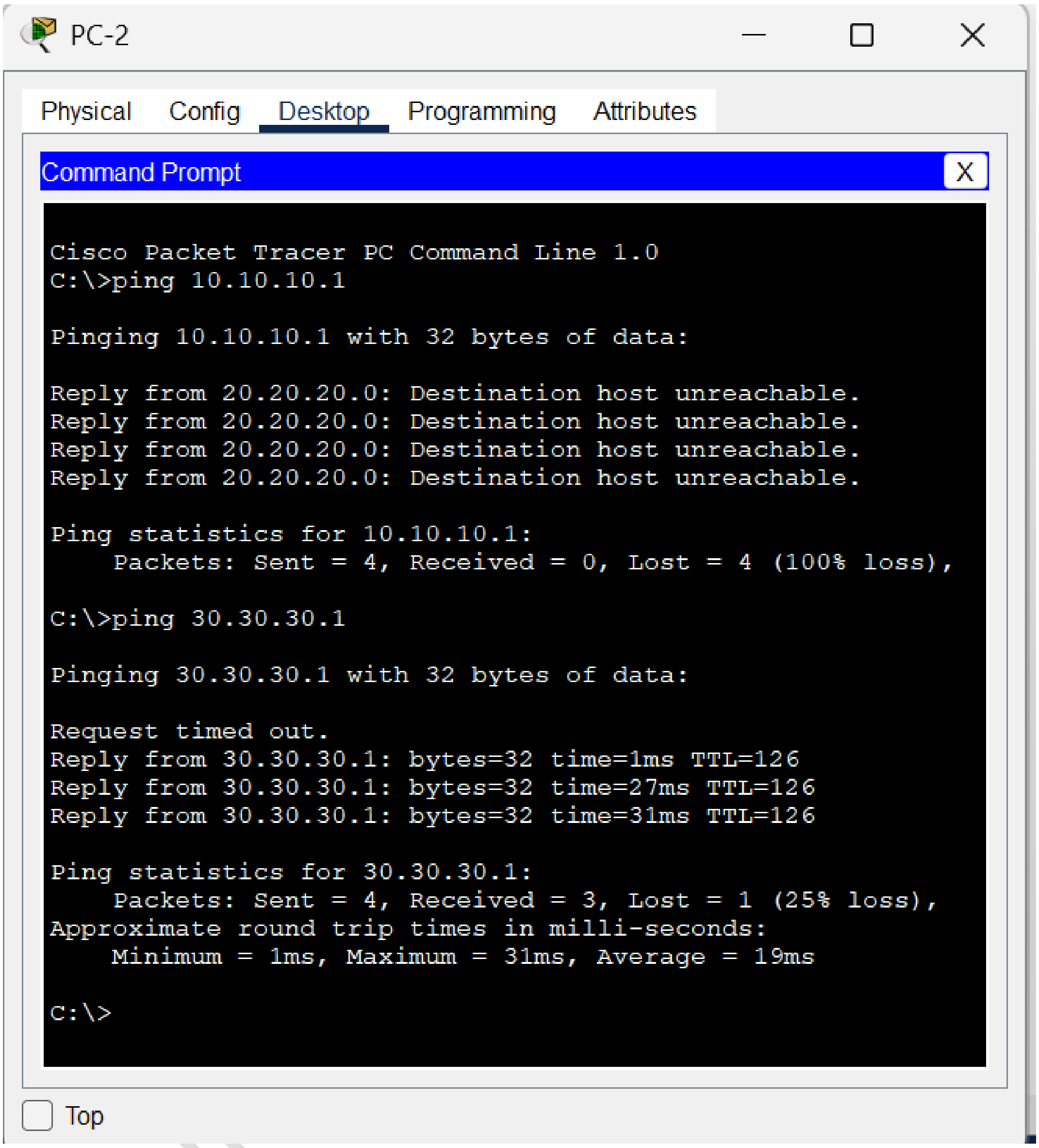
Ping test:

- Open command prompt in PC3 and ping PC2.
- Ping successful.



- Open command terminal in PC2 and ping PC3.

- Ping Failed.



Result:

Thus Network Address Translation (NAT) was examined using Cisco Packet Tracer.